

Open Source Usability Defect Classification (OSUDC)

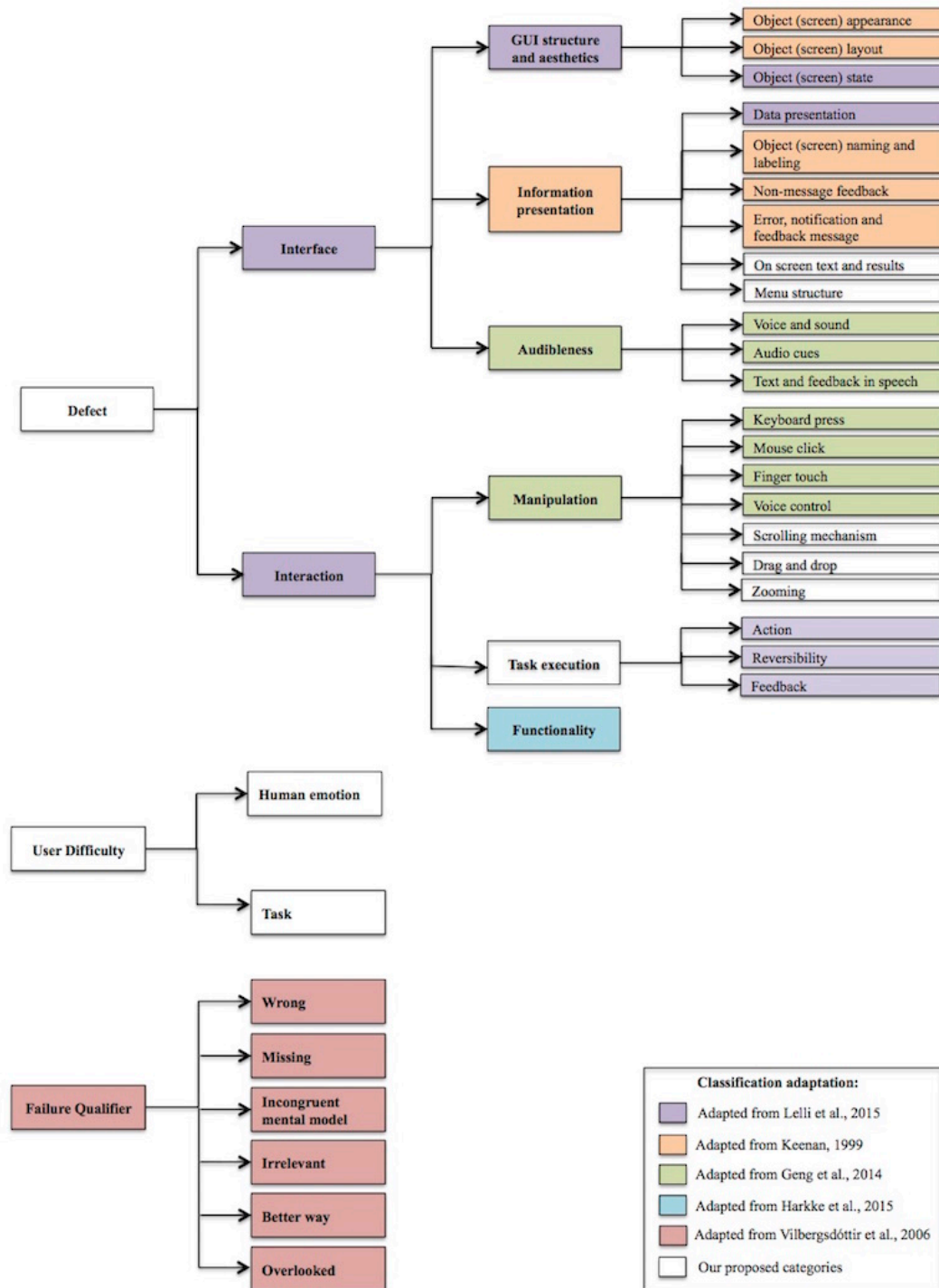


Figure 1. Hierarchical structure of usability defect categories, failure qualifier, and user difficulty attributes

Usability Defect Categories

Usability defect categories attributes contains an interface defects and interaction defects. We categorize the user *interface* defects as difficulty to view, read and hear user interface elements. We further break down user interface defects into three main categories: the *GUI structure and aesthetics*, *information presentation* and *audibleness*. Table 1 and Table 2 present definitions of these categories and samples defect phrase.

- 1) *GUI Structure and Aesthetics*: This defect category corresponds to the unexpected design, organization, and behavior of the graphical user interface (GUI). This main category contains three subcategories:
 - a. *object (screen) appearance* that concerns aesthetics aspects of GUI object (button, menu, scroll bar, icon) such as color, size and animation
 - b. *object (screen) layout* that concerns organizations of elements on the interface/ screen such as screen clutter and object proximity
 - c. *object (screen) state* that concerns on the dynamic behavior of object state, such as visible, enabled, or selected
- 2) *Information presentation*: This defect category corresponds to the visual aspects of information display on GUI and when the user has trouble to understand the words that are used in the user interface. This main category contains six subcategories:
 - a. *Data presentation* that concerns on unexpected data appearance, data properties and data type and format
 - b. *Object (screen) naming and labeling* that concerns on the use of inappropriate words on objects (such as buttons, title bars and field labels)
 - c. *Non-message feedback* that are due to distracting, annoying and confusing feedback and inappropriate use of visual cues
 - d. *Error, notification and feedback message* that concerns on the use of phrases and sentences in error message, notification message and feedback message
 - e. *On screen text and result* that concerns on the information relevancy and credibility of on screen text, instruction, online help and results of user queries
 - f. *Menu structure* that concerns on organization of menu and options
- 3) *Audibleness*: This defect category concern aspects of problem related to the way user understands sound commands and alert. This main category contains three subcategories:
 - a. *Voice and sound* concerns on distracting, disturbing and annoying sounds, or missing sound alerts when the message comes to the screen
 - b. *Text and feedback* in speech concerns on difficult to understand speech signs and translating these to text

The *interaction* defects refer to defects affecting the interaction process when a user interacts with a GUI. As shown in Figure 2, interaction defects includes three primary categories: manipulation, task execution and functionality.

- 1) *Manipulation* focus on the aspect of manipulating objects on the user interface. This type of problem occurs when the user has trouble to use the mouse (i.e., speed of cursor tracking triple clicking, depressing multiple mouse button simultaneously), keyboard (i.e., access key management), touch screen, voice control, scrolling mechanism, drag and drop and zooming
- 2) *Task execution* refers to defects that complicate a user from completing an intended action
 - a. *Action* defects refer to that produced after executing certain actions. For instance, the expected action is executed but its results are not correct. Other defects that groups under this category concerns the absence of action when interacting with the system, and execution of wrong actions
 - b. *Reversibility* concerns on the incorrect behavior of the undo/ redo operations including the interruption of action in progress
 - c. *Feedback* concerns on the absence of feedback provided by the system when the user perform critical actions, and continuous feedback to reflect the current state of an action in progress
- 3) *Functionality* refers to any problem that is due to the facilities provided by the product to user such as missing, incomplete, preference, inadequate and problematic functionality

Usability Defect Categories Definition

Defect	Definition
Interface	Any unpleasant graphical user interface aspects that affect the overall look and feel of the application
GUI structure and aesthetic	Any difficulty to read and understand the user interface objects (icon, menu item, scroll bar, button, favicon), symbol, and images.
Object (screen) appearance	Refers to how individual object look, sound, or appear to other senses. These problems involve object affordances such as the use of colours, size, and animation.
Object (screen) layout	Refers to layout coherence and how user interface objects are laid out on the screen. These problems involve spatial organization, such as the use of balance and symmetry, the alignment and spacing of elements the grouping of related elements, the placement of screen objects and consistent use of the GUI elements across applications.
Object (screen) state	Any difficulty encountered by the user when they cannot recognize or are unclear about the effect of object state change, including the change to its behaviour and appearance.
Information presentation	Any difficulty encountered by the user when they view, read, and interpret the information or data presented in the user interface.
Data presentation	Refers to how data is presented, structured and controlled.
Object (screen) naming and labelling	Any difficulty in language such as words/ terminology used such as names of objects (such as buttons, title bars, field labels) and screens [3]. These problems also include inconsistencies of naming and labelling standard.
Non-message feedback	Any difficulty that is due to distracting, annoying, and confusing feedback [3], and insignificant use of visual cues that appears while using user interface.
Error, notification and feedback message	Any difficulty in language such as words used in phrases and sentences in error, notification and feedback message. These problems also include the ability of users to understand and interpret the meaning of information presented in the message.
On screen text and results	Refers to the completeness, accuracy and credibility of information in on screen text/ instructions, online help and tutorials and results of user queries.
Menu structure	Refers to organization of menus, and grouping of related options.
Audibleness	Any unpleasant or undesirable characteristics of the audio like sound management and sound alerts.
Voice and sound	Refers to any problem related to audio cues at the interface like giving distracting, disturbing, and annoying sounds, or missing sound alerts when the message comes to the screen.
Text and feedback in speech	Refers to any problem when the interface has a difficulty to understand speech signs and translate these into text.
Interaction	Any difficulty encountered by the user when they interact with the application or while performing a task.
Manipulation	Any defects that occur when the user has trouble with some aspects of manipulating objects on the user interface.
Keyboard press	Any difficulty encountered by the user when they use the keyboard to interact with the user interface. These problems include the problematic use of access keys such as a shortcut to issue menu commands.
Mouse click	Any difficulty that involves the user's ability to use the mouse and its buttons to directly manipulate objects (speed of cursor tracking triple clicking, depressing multiple mouse buttons simultaneously) [3]
Finger touch	Any difficulty encountered by the users when they touch areas of the screen to move the pointer, press a button and manipulates an image.
Voice control	Any difficulty encountered by the users when they use voice signals to activate the user interface or invoke certain tasks.
Scrolling mechanism	Any difficulty encountered by the users when they use vertical scrollbars to moving up and down data, and horizontal scrollbars to move the data left and right within the view.
Drag and drop	Any difficulty encountered by the users when they select and drag an object or hen they drop it into another location in the interface.
Zooming	Any difficulty to change gradual image scaling operation.
Task execution	Any defects encountered by the user that inhibits the user from completing an intended action.
Action	Any defects that occur as a result of executing a task.
Reversibility	Refers to the ability of the application to allow a user to explore the interface and make mistakes and roll back the action, such as multi-level undo and the ability to cancel long-running actions.
Feedback	Refers to the ability of the application to always keep users informed about what is going on for every user event, such as prompt a warning, status and error message.
Functionality	Any problem that is due to the facilities provided by the product to the user.

Example of Primary and Subcategories of usability defects

Categories	Sub-categories	Defects	Sample defects
Interface	GUI structure and aesthetics	Object appearance (icon, menu item, scroll bar, button, favicon etc)	Object (screen) keep blinking/ flashing/ flickering
			Unnecessary animation on object (screen)
			Incorrect/ unnecessary use of object icon
			Incorrect/ inconsistent object design
			Inappropriate use of object size
			Missing object
			Unwanted object
			Incorrect/ inconsistent use of object color
			Inappropriate use of color contrast between object and background
			Inappropriate object (screen) resolution
			Object (screen) is truncated/ chopped/ overlapped/ blocked/ cropped
			Object (screen) not persistently display
			Object (screen) looks very bare
			Words/ name on object are not completely visible
		Object (screen) layout	Improper use of screen width and length
			Inconsistent margins on screen
			Limited space for displaying object
			Inconsistent/ incorrect object (screen) placement
			Object (screen) not resizable/stretchable
			Object (screen) do not automatically resize to fit in content
			Object (screen) is misaligned
			Unnecessary screen
			Object (screen) blocking the page
			Object (screen) cannot reposition/ reordered
			Incorrect screen maximized/resized behavior
			Incorrect hidden/ unhidden object (screen) behavior
		Object (screen) state	Incorrect object (screen) activation and deactivation behavior
			Indistinguishable between active and inactive object (screen)
			Incorrect object (screen) select / deselect behavior
			Incorrect object (screen) visibility behavior
			Incorrect/ unclear current object (screen) state
			Slow object state change
			Slow screen refresh
			Screen do not automatically load content/ refresh
			Cursor loss focus
			Incorrect object (screen) focus
			No visual cue to screen (object) existence
			Object action is not properly set
			Inappropriate/ incorrect screen navigational
	Information presentation	Data presentation	Missing default data
			Data not selected
			Object does not remember last selection
			Data is not populated in the object (screen)
			Incorrect data type and format
		Object (screen) naming and labeling	Misspelled/ mislabeled label
			Label name is not intuitive
			Incorrect use of labeling standard
		Non-message feedback	Inconsistent menu labeling format
			Missing visual cues
			Misunderstanding of visual cues
		Error/ notification and feedback message	Irritating/ unnecessary animation on the progress bar
			Incorrect/ missing/ non-informative indicator
			No warning indicator in the screen
		On screen text and results	Inappropriate use of dialog feedback
			Inconsistent message
			Inappropriate/ irrelevant message
		Menu structure	Incorrect/ misleading/ insufficient message
			Message is too verbose/ non -informative
			Missing/ incomplete/ misleading information
			Content is not updated
			Unsupported format to display content
	Audibleness	Voice and sounds	Irrelevant of main options to sub-options
			Inappropriate menu option
		Text/ feedback in speech	Unorganized menu organization/ structure
			Obsolete menu option
			Missing relevant menu option
			Misplaced menu option
			Inappropriate video/ audio behavior
			NS

Categories	Sub-categories	Defects	Sample defects	
Interaction	Manipulation	Keyboard press	Cursor losses focus when typing Duplicate access keys Inappropriate use of access keys Inconsistent access keys Missing/ not working access keys Unable to navigate form one field to another using keyboard Unable to type Unsupported keyboard function	
		Mouse click	Mouse click not working Mouse hover does not give focus Inconsistent mouse click behavior Too many clicking to perform task	
		Finger touch	Inappropriate vibrating behavior Incorrect click event action Tap did not invoke action bar	
		Voice control	NS	
		Scrolling mechanism	Scrolling not working Inappropriate scrolling behavior Unable to scroll In appropriate use of vertical/ horizontal scroll Unbreakable scrolling axis lock	
		Drag and drop	Unable to drag screen (object)	
		Zooming	Incorrect/ inappropriate zooming behavior Unclear visual cues that an editor is zoomed	
		Functionality	Missing Inadequate Preference Irrelevant Misaligned Problematic Technical deficiency	
		Task execution	Action	Incorrect action results No action executed Incorrect action executed Too many steps to task completion Unnecessary action to execute task Limited accessibility to execute action
				Reversibility
	Feedback			Missing notification/ warning/ feedback message No summary of task in progress/ completed Unnecessary confirmation dialogue box/ indicator/ feedback Missing permission request to access other device No indicator on the current state of an action in progress No indicator on what will be happened when clicking on certain button/ menu/option No feedback when performing critical action

** NS – No sample defects from the 377 open source usability defects we studied*